

Evaluation Protocol, Checkpointing, and Calibration Shape Apparent Multi-Task Gains in ADMET Prediction

Yuchen Fan^{1*}

¹School of Automation, Beijing Institute of Technology, Beijing, 100081,
China.

Corresponding author(s). E-mail(s): functionhx@gmail.com;

Abstract

We examine whether apparent multi-task gains in ADMET prediction are robust to evaluation protocol, probability calibration, and controls for cross-task leakage and training design. In a leakage-controlled diagnostic on eight binary ADMET endpoints from the Therapeutics Data Commons, we compare multi-task (MTL) with architecture-matched single-task (STL) models under global three-way molecule allocation (70%/10%/20% train/calibration/test) under grouped-random and scaffold-grouped protocols, with five split instances and three seeds. Under the prospectively frozen fixed-final checkpoint rule, MTL assigns lower raw log loss on all eight endpoints under both protocols, with a larger contrast under scaffold-grouped evaluation (protocol contrast $\Gamma = +0.081$, split-instance bootstrap 95% CI [$+0.048, +0.112$]). A post hoc validation-selected checkpoint sensitivity reverses the absolute MTL-STL contrast under both protocols but preserves a positive protocol interaction (split-level Γ positive in all three split instances). Strict per-model temperature calibration—each temperature fitted on the same trained model’s own calibration partition and applied to its own test partition—attenuates the raw-probability advantage and the protocol interaction to near-zero point estimates, indicating that the observed gains primarily reflect temperature-correctable differences in raw logit scale. Compute-matched, label-permutation, and pooled-pretraining controls indicate that the raw gain is not reproduced by additional single-task updates and does

not require genuine auxiliary-label structure, and the protocol interaction survives full label permutation. The positive direction of the protocol interaction was reproduced in a matched-scale graph neural network sensitivity analysis, and no systematic novelty dependence was found under either protocol. Reported multi-task advantages are therefore conditional on evaluation protocol and should be accompanied by calibration and training-rule diagnostics.

Scientific Contribution: We introduce a leakage-controlled framework for quantifying protocol-dependent MTL-STL proper-score contrasts in ADMET prediction and show that the apparent gain is larger under scaffold-grouped evaluation but attenuates to near-zero after strict per-model temperature calibration. Compute-matched and label-permutation controls further show that the raw gain is not reproduced by additional single-task updates and should not be interpreted as evidence of semantic cross-task label transfer.

Keywords: multi-task learning, ADMET prediction, evaluation protocol, calibration, benchmarking

1 Introduction

Multi-task learning (MTL) is widely used in ADMET (absorption, distribution, metabolism, excretion, and toxicity) modeling, and several studies report that multi-task models improve over matched single-task (STL) baselines [Jiang et al. \(2025\)](#); [Shahid et al. \(2025\)](#). At the same time, the molecular machine-learning community has accumulated evidence that evaluation protocols materially shape conclusions: out-of-distribution (OOD) split protocols such as scaffold or cluster splits change absolute performance and its rankings [Fooladi et al. \(2025\)](#); [Guo et al. \(2025\)](#), and recent benchmarks in neighboring fields have shown that strong model claims can depend on evaluation details [Ahlmann-Eltze et al. \(2025\)](#); [Wei et al. \(2026\)](#).

Prior studies largely report MTL-STL comparisons under a single split protocol. It remains unclear whether the paired advantage itself changes across protocols, whether proper-score gains survive model-specific recalibration, and whether those gains require genuine auxiliary-task label information rather than shared optimization and auxiliary-input exposure. To our knowledge, prior work has not jointly characterized the paired MTL-STL contrast under leakage-controlled global molecule

partitions, strict same-model recalibration, and mechanistic controls of the training design.

In this paper we report a leakage-controlled diagnostic whose final confirmatory plan was frozen before the corrected three-way rerun of task-balanced hard-sharing MTL on eight binary ADMET endpoints from the Therapeutics Data Commons [Huang et al. \(2021\)](#). The contributions are: (i) a paired MTL–STL protocol-contrast estimand evaluated under global molecule allocation with no cross-task identity exposure, under two valid protocols (grouped random and scaffold-grouped); (ii) a strict same-model calibration decomposition showing that the raw proper-score gain is attenuated to near-zero by per-model temperature rescaling; (iii) compute-matched (STL-8 $\times$), label-permuted, and pooled-pretraining controls that narrow the mechanism to concurrent shared-trunk optimization with auxiliary inputs rather than semantic label transfer or additional target-task updates; and (iv) a matched-scale GNN sensitivity analysis together with a split-instance adequacy analysis. Endpoint-scale manipulation (downsampling) and the continuous novelty analysis are reported as secondary analyses.

The paper is organized as follows. Section 2 positions the work against prior studies of ADMET MTL, split protocols, and benchmark reliability. Section 3 defines the estimand hierarchy and the leakage-controlled design. Section 4 reports the protocol contrast, the calibration decomposition, the mechanistic controls, and the robustness analyses in that order. Section 5 discusses implications and limitations.

2 Related Work

Multi-task ADMET modeling.

Multi-task architectures are common in ADMET prediction. MolP-PC [Jiang et al. \(2025\)](#) trains a multi-view, multi-task framework over 54 endpoints and reports

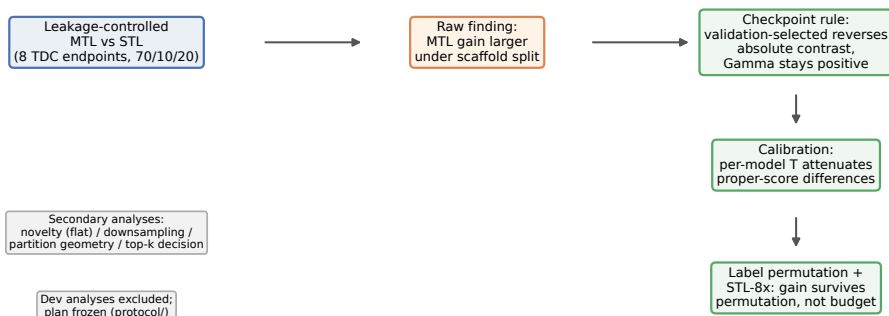


Fig. 1 Study design and principal diagnostic findings. Main evidential chain (top): leakage-controlled MTL vs. STL comparison reveals a raw gain that is larger under scaffold-grouped evaluation; diagnostic 1 (strict same-model temperature calibration) attenuates the gain to near zero; diagnostic 2 (label-permuted MTL and compute-matched STL-8 $\times$) shows the gain survives full label permutation and is not reproduced by additional single-task updates; a matched-scale GIN sensitivity reproduces the positive direction. Secondary analyses (grey): novelty dependence, downsampling, partition-geometry reweighting, and decision-level analysis.

that multi-task learning outperforms single-task training on 41 of 54 tasks; MTAN-ADMET [Shahid et al. \(2025\)](#) proposes an adaptive multi-task network over 24 endpoints with task-specific learning dynamics; industrial-scale studies report benefits of multi-task models on time-split data [Walter et al. \(2024\)](#). Reported gains vary across datasets and settings, and task balancing and endpoint size are themselves known to affect transfer in general MTL [Kearnes et al. \(2016\)](#). These studies report average or per-task performance under a single evaluation protocol and do not condition the reported contrast on protocol choice, endpoint scale, or novelty.

Split protocols and out-of-distribution evaluation.

A growing line of work examines how evaluation protocols change conclusions in molecular property prediction. [Fooladi et al. \(2025\)](#) compare 14 model classes across ten split protocols on eight datasets and find that scaffold splits are not uniformly harder than random splits while cluster-based splits are; [Guo et al. \(2025\)](#) reach similar conclusions on cancer cell lines and show that rank correlations between in-distribution and out-of-distribution performance collapse under cluster splits. [Zheng et al. \(2026\)](#) show

that structural-frontier splits expose failures hidden by scaffold splits, and Guo and Ding (2026) find that model-size conclusions depend on protocol difficulty. These studies evaluate single-task models; the paired MTL–STL contrast is not their dependent variable, and none manipulates endpoint size or measures continuous target-relative novelty.

Multi-task transfer under realistic splits.

Studies at the intersection of MTL and split protocols are sparse and reach differing conclusions. Kearnes et al. (2016) observed that multi-task gains were modest and dataset-dependent under temporal splits in industrial ADMET data; Wu et al. (2026) found negative transfer in a two-task setting under scaffold splits, mitigated by safe scheduling; Walter et al. (2024) and CheMLT-F Mun and Fazli (2026) report multi-task benefits persisting under realistic splits. These results are not directly comparable because endpoint sets, architectures, metrics, and molecule-holdout schemes differ. A compact comparison table of these studies on six axes—architecture-matched STL baseline, split protocols, global molecule holdout, endpoint-size manipulation, continuous novelty, and probabilistic calibration—appears in the supplementary material; the table makes the gap concrete: no prior study jointly varies protocol, endpoint scale, and novelty while globally partitioning molecules and reporting calibration-level metrics.

Benchmark reliability and evaluation variability.

The molecular machine-learning community has been repeatedly reminded that evaluation choices can dominate model differences. In single-cell genomics, two major benchmarks reached contradictory conclusions about foundation-model perturbation prediction Ahlmann-Eltze et al. (2025); Wei et al. (2026), with the discrepancy traced to evaluation protocol details. Within molecular property prediction, rank correlations between in-distribution and out-of-distribution performance can collapse under cluster

splits Guo et al. (2025), which implies that conclusions drawn from a single benchmark realization—including a single split instance and seed—should be treated cautiously. Our study follows this diagnostic tradition: we measure the conditions under which a design choice (task-balanced hard-sharing MTL) delivers or fails to deliver its reported advantage, with multiple split instances and seeds so that evaluation variability is quantified rather than assumed away.

3 Methods

3.1 Estimand hierarchy

For endpoint e , protocol p , split instance k , training seed r , and test set T_{epk} , let $\hat{p}_{iepk}^{\text{STL}}$ and $\hat{p}_{iepk}^{\text{MTL}}$ denote the predicted probabilities of the single-task and multi-task models for molecule i . We define the molecule-level log-loss contrast

$$d_{iepk} = \ell(y_i, \hat{p}_{iepk}^{\text{STL}}) - \ell(y_i, \hat{p}_{iepk}^{\text{MTL}}), \quad (1)$$

where $\ell(y, p) = -y \log p - (1 - y) \log(1 - p)$; positive d favors the multi-task model.

Endpoint-level and protocol-level aggregates are

$$\Delta_{epkr} = \frac{1}{|T_{epk}|} \sum_{i \in T_{epk}} d_{iepk}, \quad \Delta_p = \frac{1}{8} \sum_e \mathbb{E}_{k,r}[\Delta_{epkr}], \quad \Gamma = \Delta_{\text{scaffold}} - \Delta_{\text{random}}. \quad (2)$$

Δ_p is the primary endpoint-macro estimand (endpoints weighted equally), and Γ is the protocol-contrast estimand. We treat the eight endpoints as a fixed target set; we do not claim a superpopulation of all ADMET endpoints. Log loss is a strictly proper measure of probabilistic accuracy; ranking discrimination (AUROC/AUPRC) is assessed separately.

3.2 Models and training

The multi-task model comprises a shared 1024–256–128 encoder (ReLU activations, dropout 0.2 after the first hidden layer) and eight independent affine heads mapping the 128-dimensional representation to one logit per endpoint. Model inputs are fixed-length binary fingerprints computed with RDKit’s Morgan algorithm (`GetMorganFingerprintAsBitVect` [RDKit contributors \(2024\)](#)): radius 2, 1024 bits, default settings (`useChirality=False`, `useFeatures=False`), i.e., the standard non-chiral variant commonly denoted ECFP4; fingerprints were computed once per endpoint from the canonical SMILES and reused across all split instances and seeds. Each single-task model is an independently initialized encoder of the same architecture with one corresponding affine head; we therefore compare against *architecture-matched* single-task models, noting that capacity is matched only on the target-task pathway, not globally.

Training uses *equal task sampling with equal loss coefficients*. Each epoch consists of 64 MTL optimizer updates: eight batches of size $B = 512$ drawn with replacement from each of the eight tasks, each update using that task’s unweighted binary cross-entropy; uniform scheduling gives equal task weighting in expectation. With 60 epochs, each task’s head receives $U = 480$ updates, and the shared trunk receives 3,840 updates in total. Each single-task model receives $U = 480$ updates with $B = 512$ examples under the same sampler; collectively the eight STL models also receive 3,840 updates. Target-task exposure and optimizer-step count are therefore matched between MTL and STL; total examples processed differ (the MTL model additionally consumes auxiliary-task data). Optimization uses Adam ($\eta = 10^{-3}$, weight decay 10^{-4}) with the learning-rate schedule indexed by optimizer step. We scope all conclusions to this task-balanced hard-sharing architecture. As a sensitivity analysis, the protocol contrast was re-estimated with a graph neural network: a three-layer graph isomorphism network (GIN) with hidden width 64, sum pooling (`global_add_pool`), and per-endpoint linear

heads, trained on molecular graphs with the same task-balanced schedule (Adam, $\eta = 10^{-3}$, batch size 128, 80 epochs, eight updates per task per epoch) under the same leakage-controlled design and both protocols, at the same scale as the MLP (five split instances $\times$ three seeds). All experiments ran under Python 3.10 with PyTorch 2.13.0 (CUDA 13.0 build) on a single NVIDIA RTX 4070 Ti SUPER GPU; RDKit 2023.09.6 was used for canonicalization, fingerprinting, and scaffold computation.

3.3 Mechanistic controls

Three additional training configurations were evaluated as post hoc mechanistic controls. The label-permuted and pooled-pretraining controls were run under the random protocol at three split instances $\times$ three seeds with all other frozen settings unchanged. (i) *Label-permuted MTL*: for each target endpoint separately, the labels of all other seven endpoints are permuted within each auxiliary endpoint (fixed per-(instance, seed, endpoint) permutations), preserving input exposure, supervision count, and parameterization while removing genuine cross-task label information from the shared trunk; the target-task head still receives true labels. (ii) *Pooled pretrain + STL fine-tune*: a hard-sharing MTL run is used as a shared-trunk initialization, after which each endpoint’s head is re-initialized and the full network is fine-tuned on that endpoint alone with the same task-balanced update schedule. These controls compare (i) label transfer against data exposure and (ii) joint-training gains against transferable-representation gains. A further compute-matched control (STL-8 $\times$) trains each single-task model on its own true data alone but with the multi-task trunk’s full 3,840 optimizer updates (identical batch size, optimizer, and step-indexed schedule), so that optimization budget is matched while auxiliary-task input and labels are absent. The protocol-scale matrix is: label-permuted MTL and pooled pretraining were evaluated under grouped-random allocation at three split instances $\times$ three seeds; the compute-matched STL-8 $\times$ control was evaluated under grouped-random allocation at

five instances $\times$ three seeds; the label-permuted MTL and STL-8 $\times$ controls were additionally repeated under scaffold-grouped allocation at three instances $\times$ three seeds; pooled pretraining was evaluated only under grouped-random allocation. All controls are post hoc mechanistic diagnostics reported as such; the confirmatory contrasts are unaffected.

3.4 Leakage-controlled evaluation

A test molecule held out from endpoint e may still appear in the training partition of another endpoint, exposing its structure to the multi-task model through auxiliary labels. We control for this *cross-task identity exposure* by global molecule allocation: each molecule is assigned to a single global partition, and every endpoint inherits that partition. Evaluation is therefore inductive with respect to molecule identity across tasks: a test molecule cannot appear in any auxiliary endpoint’s training partition. No cross-task identity exposure remains under this design. Molecule identity is canonicalized by RDKit canonical-SMILES round-trip (`MolFromSmiles/MolToSmiles`) under a fixed RDKit version, which retains defined stereochemistry in the canonical form. Per endpoint, rows that fail canonicalization are dropped, and rows sharing a canonical SMILES are deduplicated keeping the first occurrence; this is the frozen rule for duplicate and conflicting labels, applied identically in the development and confirmatory pipelines. Across the eight endpoints, 13 molecules carry conflicting labels (3 in hERG, 10 in BBB_Martins; none elsewhere); excluding all of them changes the endpoint-macro contrasts by less than 2×10^{-4} under both protocols and both raw and calibrated variants, so the deduplication rule does not drive any reported result. We evaluate two protocols under this scheme: grouped random allocation and Bemis–Murcko scaffold-grouped allocation [Bemis and Murcko \(1996\)](#) (with a defined policy for acyclic molecules and empty scaffolds). Five split instances and three training seeds are used; split and seed lists, and their hashes, are frozen. Each instance allocates

70% of the unique-molecule pool to training, 10% to calibration, and 20% to test, and this allocation is global: a molecule’s partition is identical across all endpoints. The calibration partition exists solely for per-model temperature fitting (Section 3.7) and is disjoint from both training and test within each instance. The confirmatory dataset comprises 33,238 unique molecule–endpoint pairs from 17,972 unique canonical molecules; across the 5×3 (instance, seed) combinations this yields 148,008 test observations, with each (instance, seed) combination containing 9,814–9,930 unique molecule–endpoint pairs and no within-combination duplicates. Cross-instance overlap arises because different split instances draw different test partitions; the identity key is the same canonical SMILES used for global allocation.

3.5 Continuous target-relative novelty

For each test molecule we define *target-relative novelty* as one minus the maximum Tanimoto similarity (ECFP4 [Morgan \(1965\)](#), fixed RDKit version [RDKit contributors \(2024\)](#)) to any molecule in the target endpoint’s training partition. Distance to auxiliary-task training molecules is deliberately excluded from the primary definition, so the axis measures distance from the endpoint’s own training support; we preregister a secondary quantity for nearest-auxiliary-training similarity and its interaction with target-relative novelty, since a molecule far from target training but close to auxiliary training is precisely a candidate transfer case.

3.6 Endpoint scale and downsampling

Endpoint scale is the number of unique target-task training molecules in each split. With eight endpoints, an observational scale association is weak and confounded by endpoint identity; we therefore prespecify a downsampling experiment: large endpoints are subsampled to a fixed target-training size with a frozen sampler (nested subsets, class prevalence preserved, fixed validation and test partitions, auxiliary-task data

unchanged, task-balanced schedule held constant). The defensible conclusion from this experiment is an effect of *target-label availability* within the examined endpoints under this training schedule, not a general “scale rather than identity” claim.

3.7 Calibration and discrimination decomposition

A Brier decomposition explains the Brier-score contrast, not the log-loss contrast. We therefore report (i) a Brier reliability–resolution decomposition as a descriptive complement, (ii) calibration intercept and slope, and (iii) AUROC/AUPRC contrasts. A finding of stable AUROC together with worse Brier reliability is reported as evidence *consistent with* a calibration difference, not as proof that the NLL penalty is primarily calibration.

Temperatures are fitted per run, protocol, model, and endpoint, and strictly within the same trained model. Global molecule allocation now assigns each molecule to one of three partitions: 70% training, 10% calibration, and 20% test (random protocol: molecule-level; scaffold protocol: scaffold-grouped). For each (split instance, training seed) run, one scalar temperature T is fitted by minimizing the mean log loss (NLL) on that same trained model’s own calibration partition (disjoint from its training and test partitions; bounded scalar optimization, SciPy `minimize_scalar`, method `bounded`, $T \in [0.2, 5.0]$) and applied to that same model’s own test partition; no temperature is shared across models. The calibrated molecule-level contrast is recomputed from temperature-scaled probabilities.

3.8 Checkpoint-selection sensitivity

As a post hoc sensitivity analysis, the calibration partition (10%) is split molecule-level into validation (5%) and temperature-calibration (5%) subsets under each protocol (scaffold grouping applies to the allocation of the parent calibration partition). During training, validation NLL is evaluated at every epoch (the mean over the eight heads for

MTL, each head on its own endpoint’s validation partition; the endpoint’s own validation partition for STL), and the best-validation checkpoint is retained. Temperature is then fitted on the 5% temperature-calibration subset of the same trained model and applied to its test predictions. Two symmetric selection rules are additionally computed: global-vs-global (the eight STL models jointly select the epoch minimizing the endpoint-macro validation NLL, matching the single MTL selection) and endpoint-vs-endpoint (each MTL head and its corresponding STL select on that endpoint’s own validation NLL). The analysis is reported descriptively at the split level because the three seeds within a split share a test set.

3.9 Uncertainty and preregistration

The fifteen split-seed combinations are not fifteen independent replicates: three training seeds share a test set, split instances reuse molecules, all endpoints share one MTL model, and a molecule can carry multiple endpoint labels. Confirmatory inference therefore uses paired contrasts with clustering by standardized molecule, distinguishing split variability from training variability; we report split-level and seed-level dispersion separately. The analysis plan—hypotheses, directional tests, split and seed hashes, estimands and weighting, exclusions, downsampling design, novelty model, multiplicity handling, and analysis code version—was frozen in an immutable record (repository commit 5c2cd1e; protocol/final_confirmatory_plan.md) before confirmatory computation on the leakage-controlled design; the development pass (a per-endpoint split subsequently identified as cross-task-exposed) informed the design and is reported separately as a contaminated stress test (Additional file 1, Section S2), excluded from confirmatory statistics.

Endpoint	Test predictions	Unique molecules	Positive rate
hERG	1,938	443	0.69
AMES	21,933	4,914	0.55
BBB_Martins	5,907	1,322	0.76
Pgp_Broccatelli	3,747	832	0.53
CYP2C9_Veith	36,201	8,132	0.33
CYP2D6_Veith	39,246	8,829	0.19
CYP3A4_Veith	37,161	8,350	0.41
Bioavailability_Ma	1,875	416	0.77

Table 1 Dataset statistics. “Test predictions” counts molecule–endpoint test observations across 5 split instances $\times$ 3 seeds (within-(instance, seed) sets contain no duplicates; unique molecules across instances).

4 Results

We report results for eight binary ADMET endpoints from the Therapeutics Data Commons [Huang et al. \(2021\)](#): hERG, AMES, BBB penetration, P-glycoprotein inhibition, three cytochrome P450 inhibition tasks, and oral bioavailability (Table 1). All conclusions below are restricted to this endpoint set and to task-balanced hard-sharing MTL.

Under the prospectively frozen fixed-final checkpoint rule, across the leakage-controlled design (global molecule allocation, five split instances, three seeds), the molecule-level log-loss contrast Δ is positive for every endpoint under both protocols: multi-task models assign lower log loss than their architecture-matched single-task counterparts in all 8×2 protocol–endpoint configurations.

4.1 Protocol interaction and overall effect

The endpoint-macro mean contrast is $\Delta_{\text{random}} = +0.321$ under grouped random allocation and $\Delta_{\text{scaffold}} = +0.402$ under Bemis–Murcko scaffold-grouped allocation, with a protocol contrast of $\Gamma = \Delta_{\text{scaffold}} - \Delta_{\text{random}} = +0.081$ (paired $t_7 = 2.65$, $p = 0.033$). Because the eight endpoints and five split instances are fixed in this study, uncertainty in Γ has three distinct components, which we report separately. Conditional on the observed split instances and training runs, molecule-level resampling gives a

narrow interval (95% CI $[+0.079, +0.083]$); this answers only "how much sampling
 noise among molecules," not variability across instance realizations. A two-level block
 bootstrap that resamples split instances first (then aggregates endpoints within them)
 gives 95% CI $[+0.048, +0.112]$ ($P(\Gamma > 0) = 1.00$), covering split-instance variability. An endpoint-resample bootstrap (resampling the eight Γ_e values) gives 95% CI
 $[+0.026, +0.138]$ ($P(\Gamma > 0) = 0.998$), covering endpoint heterogeneity; the across-
 endpoint standard deviation of Γ_e is 0.088. All three intervals exclude zero: the positive
 protocol contrast remained stable under molecule-, split-instance-, and endpoint-level
 resampling within this benchmark panel. The endpoint-resample interval is reported
 as a sensitivity analysis; the eight endpoints are a fixed target set, so it does not imply
 an ADMET-endpoint population. The per-endpoint contrasts Γ_e are positive in six of
 eight endpoints (Figure 2, panel B); the interaction is concentrated among the smaller
 endpoints in this panel (hERG, AMES, BBB, Bioavailability, each $\Gamma_e \approx +0.16$), while
 the two largest CYP endpoints show $\Gamma_e \approx 0$ or slightly negative (CYP2C9 -0.012 ,
 CYP3A4 -0.028). A leave-one-endpoint-out analysis leaves the interaction in the range
 $+0.069$ to $+0.096$, so no single endpoint carries the interaction. We present these
 as concordance and sensitivity analyses for the eight fixed endpoints; the endpoint-
 dispersion interval does not imply an ADMET endpoint population, and the paired
 test is a fixed-set diagnostic, not a superpopulation claim. The multi-task benefit is
 therefore *larger under the scaffold protocol* for the smaller endpoints, and the paired
 MTL-STL contrast is protocol-dependent: moving from grouped-random to scaffold-
 grouped evaluation need not shrink the observed MTL advantage. We report the
 contrast on these eight fixed endpoints; we do not claim an ADMET-wide population
 effect.

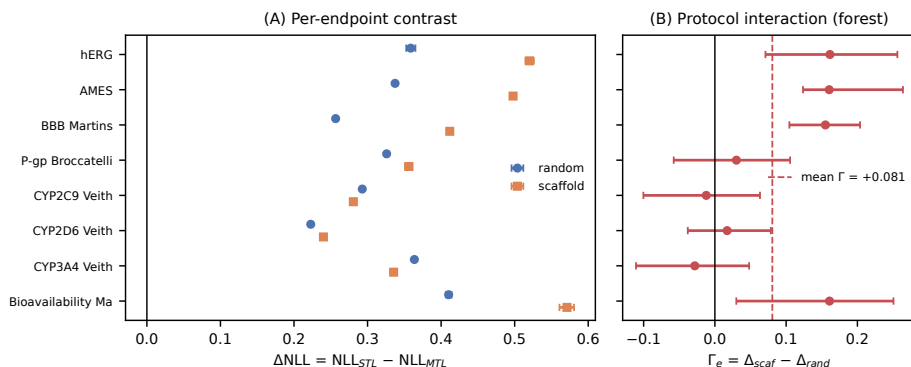


Fig. 2 (A) Per-endpoint log-loss contrast $\Delta = \text{NLL}_{\text{STL}} - \text{NLL}_{\text{MTL}}$ (positive values favor MTL) under grouped random (circles) and scaffold-grouped (squares) allocation, with molecule-level cluster-bootstrap 95% intervals. (B) Per-endpoint protocol contrasts $\Gamma_e = \Delta_{\text{scaf}} - \Delta_{\text{rand}}$ (forest; error bars: split-instance unit-level bootstrap 95% intervals) with endpoint-macro mean (dashed). Six of eight endpoints show positive Γ_e ; leave-one-endpoint-out range +0.069 to +0.096.

4.2 Calibration/discrimination decomposition and robustness

The contrast decomposes into raw-probability and calibration components. Under the random protocol, Brier scores and reliability components improve for all eight endpoints while AUROC is essentially unchanged; under the scaffold protocol, Brier and reliability improve everywhere and AUROC improves in four of eight endpoints (e.g., hERG $0.773 \rightarrow 0.812$). To test whether the log-loss advantage reflects a durable calibration difference or merely better raw probability estimates, we fitted a per-endpoint temperature T Guo et al. (2017) under the strict per-model design: for each (split instance, training seed) run, protocol, model, and endpoint, one scalar temperature was fitted by minimizing the mean log loss on that same trained model’s own calibration partition (disjoint from its training and test partitions) and applied to that same model’s own test partition; no temperature is shared across models. After this calibration, the mean contrast collapses from +0.321 to -0.003 under the random protocol and from +0.402 to +0.003 under the scaffold protocol, and the calibrated protocol contrast is $\Gamma_{\text{cal}} = +0.006$ (per-endpoint calibrated contrasts positive in five of eight endpoints; paired $t_7 = 0.80$, $p = 0.45$). The uncalibrated gains—including the protocol

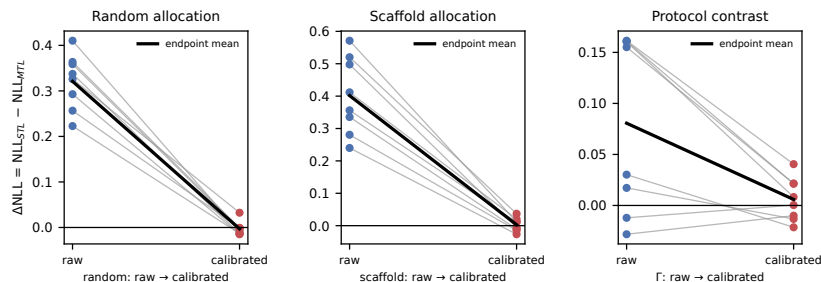


Fig. 3 Temperature-calibration contrast. Mean Δ under both protocols, raw versus strict per-model temperature calibration: the raw advantage (+0.321 random, +0.402 scaffold) attenuates to approximately zero, and the protocol contrast Γ from +0.081 to +0.006 (points: per-endpoint values).

interaction ($\Gamma_{\text{raw}} = +0.081$ vs. $\Gamma_{\text{cal}} = +0.006$)—are therefore attenuated to near-zero point estimates by per-model temperature rescaling (Figure 3), consistent with differences in raw logit scale rather than with ranking information or a calibration advantage that survives recalibration; we report point estimates and note that a formal equivalence analysis against a practical margin is needed before claiming elimination. The calibration statement is checkpoint-rule specific: under the frozen fixed-final recipe the positive raw contrasts attenuate to near zero, while under validation-selected checkpoints calibration substantially reduces but does not eliminate the negative contrasts ($-1.77 \rightarrow -0.38$ random, $-1.60 \rightarrow -0.34$ scaffold); the checkpoint rule sets the direction of the absolute contrast, calibration strongly modulates its magnitude, and the protocol interaction is positive under both rules. Class-conditional analysis shows the contrast is carried predominantly by positive-class molecules (+0.44 vs. +0.21 for negatives under random allocation).

Checkpoint-selection sensitivity. The fixed final-step recipe is part of the frozen design; we additionally asked whether the headline protocol interaction survives validation-based checkpoint selection. Splitting the calibration partition into validation (5%) and temperature-calibration (5%) subsets, selecting the best-validation checkpoint during training, and fitting temperature on the remaining calibration subset, we find that under this recipe validation NLL rises monotonically with epoch for both

MTL and STL (log loss penalizes the overconfidence that grows with training), so selection returns an early checkpoint at which the MTL shared trunk is undertrained. The absolute contrast then reverses under both protocols (-0.49 random and -0.45 scaffold with per-epoch selection, versus $+0.30/+0.31$ at the fixed final step; a coarser ten-epoch grid produces a larger reversal because it can only select the earliest evaluated checkpoint), showing that the *sign of the absolute raw contrast is checkpoint-rule dependent*. We verified that the reversal is not an artifact of asymmetric selection freedom: under global-vs-global selection (both MTL and the STL set select a common epoch by endpoint-macro validation NLL) and under endpoint-vs-endpoint selection (each head selects on its own endpoint’s validation NLL), the absolute contrasts are nearly identical (-1.35 (global) and -1.36 (endpoint) under random and $-1.26/-1.26$ under scaffold allocation, five-epoch snapshot grid). Critically, the protocol interaction is positive under every checkpoint rule examined: split-level contrasts are positive in all three split instances with per-epoch selection (mean $+0.05$), positive with the ten-epoch grid ($+0.22$), and positive under both symmetric rules ($+0.10$, two of three splits positive). The headline finding—that the paired MTL–STL contrast is larger under scaffold-grouped evaluation—therefore holds across checkpoint rules, while the magnitude (and even the sign) of the absolute contrast is a property of the frozen training recipe. We report the fixed-final rule as primary because it is the prospectively frozen recipe, and the validation-selected analyses as post hoc sensitivities (Methods); all analyses share the same train/validation/calibration/test isolation, and the sensitivity is reported descriptively at the split level because the three seeds within a split share a test set, with the 5% validation subset small for the two smallest endpoints.

4.3 What drives the contrast: mechanistic controls

The MTL and STL systems differ in representation sharing but also in data exposure and optimizer budget: the MTL shared trunk receives 3,840 optimizer updates (eight

tasks $\times$ 480 per-task updates) versus 480 for each STL trunk, and the MTL model
 processes auxiliary-task molecules the STL model never sees. Three controls under
 the random protocol isolate these channels (three split instances $\times$ three seeds for the
 label and pretraining controls; five instances $\times$ three seeds for the budget control; all
 other frozen settings unchanged). First, a *compute-matched STL* baseline (STL-8 $\times$):
 each endpoint is trained on its own true data alone but with the MTL trunk’s 3,840
 updates (identical batch size, optimizer, and schedule). STL-8 $\times$ is *worse* than stan-
 dard STL (-0.096 log-loss units; negative in all 15 runs): the multi-task advantage
 was not reproduced by extending single-task training to the matched 3,840-update
 budget under the frozen optimizer and learning-rate schedule. (We report the fixed
 final-step checkpoints; no test-based early stopping was used, so the control answers
 “same final update budget,” not “independently tuned best STL.”) Second, a *label-*
permuted MTL model: for each target endpoint separately, the labels of all *other* seven
 endpoints are permuted within each endpoint (fixed per-(instance, seed, endpoint)
 permutations; identical input exposure, supervision count, and parameterization), so
 the shared trunk cannot extract genuine cross-task label information while the tar-
 get head still receives true labels. This fully permuted control leaves the contrast
 essentially unchanged ($+0.296$ vs. $+0.298$ for standard MTL; $+0.387$ vs. STL-8 $\times$),
 indicating that the raw gain does not require genuine auxiliary-label information
 and is not reproduced by additional target-task updates. Third, a shared trunk pre-
 trained jointly on all endpoints and then fine-tuned per endpoint (head re-initialized,
 full-network fine-tuning) retains only a small fraction of the contrast ($+0.036$, ver-
 sus $+0.300$ for hard-sharing joint training), showing that most of the gain requires
 the joint training objective itself rather than a transferable shared representation
 alone (Figure 4). The same two decisive controls were repeated under the scaffold
 protocol (three split instances $\times$ three seeds): label-permuted MTL leaves the scaffold-
 protocol contrast essentially unchanged ($+0.441$ vs. $+0.407$ for standard MTL), and

the protocol interaction survives full label permutation: at matched three-instance scale, split-level paired bootstrap gives $\Gamma_{\text{real}} = +0.096$ (95% CI $[+0.058, +0.131]$) and $\Gamma_{\text{permuted}} = +0.128$ (95% CI $[+0.095, +0.163]$)—both positive with overlapping intervals, and the difference $\Gamma_{\text{permuted}} - \Gamma_{\text{real}} = +0.033$ (95% CI $[+0.006, +0.056]$, excluding zero): the interaction is not merely preserved under full label permutation but is slightly larger, though the intervals overlap. The protocol-sensitive gain is therefore not attributable to semantic cross-task label transfer. STL-8 $\times$ is again worse than STL under the scaffold protocol (-0.125), confirming that additional single-task updates do not reproduce the gain in either protocol. Together, the controls narrow the plausible explanation to effects specific to concurrent shared-trunk optimization with auxiliary inputs—including auxiliary-input exposure and gradient-noise regularization—rather than semantic cross-task label transfer or additional target-task updates; they do not uniquely identify a single mechanism. Consistent with the main calibration finding, the raw gain of every control configuration is attenuated to near zero by per-model temperature calibration (standard MTL $+0.298 \rightarrow -0.001$; label-permuted $+0.296 \rightarrow -0.016$; pooled pretraining $+0.036 \rightarrow +0.011$; within-run temperature fits on an internal calibration subset): concurrent shared-trunk training alters the logit scale even without meaningful auxiliary labels, and model-specific calibration removes most of the apparent proper-score advantage in every configuration. An earlier control that permuted only a fixed subset of auxiliary endpoints produced unstable negative transfer; we attribute that instability to optimization sensitivity of the partially permuted configuration and report only the fully permuted design here. These controls are endpoint-local diagnostics of the main finding.

4.4 Cross-architecture robustness

To test whether the protocol interaction is specific to fingerprint MLPs, we repeated the core comparison with a three-layer graph isomorphism network (GIN) on molecular

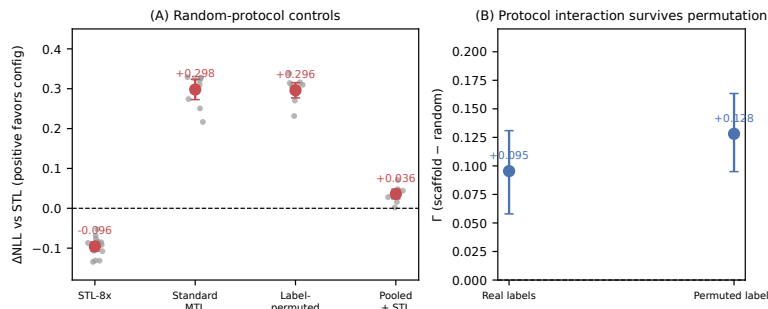


Fig. 4 (A) Mechanistic controls under grouped-random allocation (points: split-seed estimates; error bars: 95% intervals across runs). Compute-matched STL-8 $\times$ is *worse* than standard STL (-0.096), while standard MTL improves log loss over STL by $+0.298$, fully permuting all auxiliary labels (target-specific) leaves the contrast essentially unchanged ($+0.296$), and pooled pretraining followed by per-endpoint fine-tuning retains only $+0.036$. (B) Protocol interaction under real vs. permuted labels (three-instance matched scale; split-level paired bootstrap 95% intervals): $\Gamma_{\text{real}} = +0.096$ (CI $[+0.058, +0.131]$) and $\Gamma_{\text{permuted}} = +0.128$ (CI $[+0.095, +0.163]$). The protocol-sensitive gain survives full label permutation.

graphs, under the same leakage-controlled design at matched scale (five split instances, three seeds, architecture-matched STL). The positive direction of the protocol contrast is reproduced: $\Delta_{\text{random}} \approx 0$ (-0.001) and $\Delta_{\text{scaffold}} = +0.025$, giving $\Gamma_{\text{GNN}} = +0.026$ (unit-level bootstrap 95% CI $[+0.007, +0.046]$, excluding zero; a split-instance block bootstrap over the five instances gives 95% CI $[-0.006, +0.064]$ with $P(\Gamma > 0) = 0.94$, so at the instance level the GNN interaction is directionally supportive rather than individually significant), although its magnitude and endpoint pattern differ from the fingerprint MLP (positive Γ_e in four of eight endpoints, driven by hERG and Bioavailability). We read the GNN result as supporting the direction of the interaction in a second representation and model family rather than establishing representation independence (an earlier GNN run with a smaller training budget showed apparent negative transfer; this underfitting artifact and its resolution are recorded in Additional file 1, Section S10).

Endpoint-effect ordering across split instances is not stable (mean pairwise Kendall $\tau = -0.04$ under random and $+0.39$ under scaffold allocation), indicating that conclusions about which endpoints benefit most should not be drawn from a single split

realization. Full per-endpoint estimates (contrast, AUROC, AUPRC, Brier, reliability, unique molecule counts) appear in Additional file 1, Table S2.

4.5 Endpoint-scale moderation

Observationally, the contrast is largest for the smallest endpoints under both protocols (e.g., Bioavailability_Ma +0.41; hERG +0.36 under random) and smallest for the largest ones (CYP2D6 +0.22), and the same ordering holds under scaffold allocation (Figure 2). A prespecified downsampling intervention subsampled the three CYP endpoints to 2,000 target-task training molecules under the current 70/10/20 design: the contrast increased in all three (CYP2C9 +0.28 $\rightarrow$ +0.40; CYP2D6 +0.24 $\rightarrow$ +0.38; CYP3A4 +0.34 $\rightarrow$ +0.53), consistent with shared representations helping more when target-task labels are scarcer; because the trunk is shared, non-CYP endpoints also moved (mixed direction), so we do not claim a causal “scale rather than identity” mechanism (full details in Additional file 1, Section S4).

4.6 Continuous novelty analysis

Under the confirmatory design the contrast shows no systematic dependence on target-relative novelty under either protocol: within-endpoint novelty slopes straddle zero for most endpoints, and a hierarchical interaction model yields a detectable but negligible effect ($\hat{\beta} = -0.006$; 95% CI $[-0.010, -0.001]$). The preregistered transfer-window hypothesis is not supported, and an apparent reversal seen in a three-instance development analysis did not survive the five-instance design (Additional file 1, Figures S1 and S2). A partition-geometry control (reweighting random-protocol test molecules to the scaffold novelty distribution) leaves the random-protocol contrast unchanged (+0.3004 vs. +0.3006), so the scaffold advantage is not explained by the between-protocol novelty shift.

967 4.7 Decision-level implications

968
969 A top- k prioritization comparison on the three toxicity endpoints (computed within
970 each run) showed only small and heterogeneous ranking differences between MTL
971 and STL (full results in Additional file 1, Section S6); we do not claim a general
972 prioritization advantage.
973
974

980 5 Discussion

983 5.1 What the results show

984
985 The results qualify, rather than overturn, the commonly reported multi-task advan-
986 tage in ADMET modeling [Jiang et al. \(2025\)](#); [Shahid et al. \(2025\)](#): under the frozen
987 fixed-final recipe and leakage-controlled evaluation, task-balanced hard-sharing MTL
988 provides better raw probability estimates on all eight endpoints and both protocols.
989
990 The headline finding is the protocol interaction: the paired MTL–STL proper-score
991 contrast is +0.321 under random allocation and +0.402 under scaffold allocation ($\Gamma =$
992 +0.081; paired $t_7 = 2.65$, $p = 0.033$), and the positive contrast was robust to molecule-
993 , split-instance-, and endpoint-level resampling (split-instance block bootstrap 95%
994 CI [+0.048, +0.112]); leave-one-endpoint-out range +0.069 to +0.096, positive in six
995 of eight endpoints and concentrated among the smaller endpoints. Under the confir-
996 matory five-instance design we found no systematic novelty dependence of the benefit
997 under either protocol. The raw-probability advantage—and the protocol interaction
998 itself—does not survive strict per-model temperature calibration, in which each tem-
999 perature is fitted on the same trained model’s own calibration partition and applied
1000 to its own test partition (calibrated $\Gamma = +0.006$, paired $p = 0.45$), so the gains reflect
1001 temperature-correctable differences in raw logit scale rather than ranking information
1002 or a durable calibration advantage.
1003
1004
1005
1006
1007
1008
1009
1010
1011

5.2 Relation to prior split-protocol and MTL findings

The protocol-dependence finding extends single-task protocol studies [Fooladi et al. \(2025\)](#); [Guo et al. \(2025\)](#); [Zheng et al. \(2026\)](#) to the multi-task contrast, and it helps resolve the apparent tension between studies reporting multi-task benefits under realistic splits [Walter et al. \(2024\)](#); [Mun and Fazli \(2026\)](#) and studies reporting negative transfer [Wu et al. \(2026\)](#); [Kearnes et al. \(2016\)](#): the direction and magnitude of the benefit depend on protocol and on the metric (proper score versus ranking). The temperature-calibration result adds a methodological caution: raw-probability advantages that disappear under recalibration should be reported as such, and calibration diagnostics should accompany proper-score comparisons. The decision-level analysis revealed only small and heterogeneous ranking differences: MTL matched or exceeded STL in most run-level top- k comparisons, but two protocol-endpoint- k combinations favored STL more often, so we do not infer a general compound-prioritization advantage.

5.3 Methodological lessons

Two results in this study were design-sensitive in ways that reward explicit reporting. First, a three-instance scaffold analysis showed an apparent reversal of the novelty dependence that disappeared under the five-instance design; the GNN sensitivity analysis similarly showed apparent negative transfer before the training budget matched the MLP configuration. Both episodes illustrate that grouped-allocation studies should verify instance and budget adequacy before interpreting pattern-level claims, and they motivate the frozen protocol and the separation of development from confirmatory analyses adopted here. Second, the strict per-model calibration design—fitting each temperature on the same model’s own calibration partition—yields the same qualitative conclusion as an earlier cross-instance calibration shortcut, but the strict design

removes a legitimate reviewer objection to the shortcut; we recommend the strict design whenever calibration is part of the evidence.

5.4 Limitations

The endpoint set is limited to eight binary TDC endpoints and one task-balanced hard-sharing fingerprint MLP architecture; other architectures, weighting schemes, and endpoint families may behave differently. The downsampling manipulation increased the contrast consistently across the three CYP endpoints, but because the multi-task model shares its trunk, the intervention also altered the auxiliary endpoints; the result therefore supports target-label availability as a moderator within this training configuration but does not isolate a general causal sample-size mechanism. Endpoint-effect ordering is unstable across split instances, and with eight endpoints the protocol interaction, while supported by the paired test and bootstrap intervals at all three resampling levels, rests on a small number of endpoint-level observations; we do not claim an ADMET-wide population effect. The temperature-calibration analysis is a point-estimate attenuation analysis; a formal equivalence analysis against a practical margin would be needed to claim elimination, and the calibrated point estimates should be read as approximately zero rather than as proof of exact zero. The top- k decision analysis is endpoint-local and descriptive. We treat all findings as benchmark-local.

6 Conclusion

We presented a leakage-controlled diagnostic with a frozen confirmatory plan of task-balanced hard-sharing multi-task ADMET models. Under the frozen fixed-final checkpoint rule, the multi-task proper-score contrast is positive on all eight endpoints under both protocols and larger under scaffold-grouped evaluation; a validation-selected checkpoint sensitivity reverses the absolute contrast under both protocols but

preserves the positive protocol interaction under both per-epoch and ten-epoch selection grids (split-level contrasts +0.005, +0.094, +0.054). Under the fixed-final rule the contrast is $\Gamma = +0.081$ with split-instance 95% CI [+0.048, +0.112] (positive in six of eight endpoints, robust to molecule- and endpoint-level resampling), attributable to better raw probability estimates that do not survive strict per-model temperature calibration (calibrated $\Gamma = +0.006$, $p = 0.45$). Under the confirmatory five-instance design we found no systematic novelty dependence of the benefit under either protocol; an apparent reversal in a three-instance development analysis did not survive the larger design.

Compute-matched and label-permutation controls (under both protocols) indicate that the raw gain is not reproduced by additional single-task updates and does not require genuine auxiliary-label structure—the protocol interaction survives full label permutation—and the positive direction of the protocol interaction was reproduced in a matched-scale GNN sensitivity analysis ($\Gamma_{\text{GNN}} = +0.026$), though at the split-instance level the GNN interval includes zero (95% CI [−0.006, +0.064]) and we read the result as directionally supportive rather than individually significant. These results suggest that multi-task proper-score advantages in ADMET modeling should be reported conditionally on evaluation protocol and accompanied by calibration diagnostics, and that instance count in grouped-allocation studies should be large enough to stabilize group sampling. Natural next steps are to extend the diagnostic to larger endpoint panels, alternative multi-task formulations, and additional OOD protocols such as cluster or temporal splits.

1151 **Declarations**

1152

1153 **Availability of data and materials**

1154

1155 The dataset(s) supporting the conclusions of this article are available in the
1156 Therapeutics Data Commons repository (<https://tdcommons.ai/>), endpoint identi-
1157 fiers: hERG, AMES, BBB_Martins, Pgp_Broccatelli, CYP2C9_Veith, CYP2D6_Veith,
1158 CYP3A4_Veith, Bioavailability_Ma. The analysis code, global molecule-partition split
1159 manifests, prediction tables, temperature-calibration module, and the full analy-
1160 sis protocol are available in the GitHub repository [https://github.com/Functionhx/](https://github.com/Functionhx/ADMET-MTL-Diagnostic)
1161 [ADMET-MTL-Diagnostic](https://github.com/Functionhx/ADMET-MTL-Diagnostic) (MIT license; frozen submission snapshot: release v1.0.0).
1162 Archived version with persistent DOI: <https://doi.org/10.5281/zenodo.21772295>.
1163
1164
1165
1166
1167
1168

1169

1170

1171 **Competing interests**

1172

1173 The authors declare that they have no competing interests.

1174

1175

1176

1177

1178

1179

1180 This research received no specific grant from any funding agency in the public,
1181 commercial, or not-for-profit sectors.

1182

1183

1184

1185

1186

1187

1188

1189

1190

1191

1192

1193

1194

1195

1196

Not applicable.

Use of AI tools	1197
AI-assisted tools were used for writing support and code assistance during the prepa-	1198
ration of this manuscript; all experiments, analyses, and conclusions were performed	1199
and verified by the author.	1200
	1201
	1202
	1203
	1204
	1205
	1206
	1207
List of abbreviations	1208
	1209
• ADMET: Absorption, distribution, metabolism, excretion, and toxicity	1210
• AUPRC: Area under the precision-recall curve	1211
• AUROC: Area under the receiver operating characteristic curve	1212
• ECFP4: Extended connectivity fingerprint, radius 4 equivalent (Morgan radius 2)	1213
• GIN: Graph isomorphism network	1214
• MTL: Multi-task learning	1215
• NLL: Negative log likelihood	1216
• OOD: Out of distribution	1217
• STL: Single-task learning	1218
• TDC: Therapeutics Data Commons	1219
	1220
	1221
	1222
	1223
	1224
	1225
	1226
	1227
	1228
	1229
	1230
Additional file	1231
	1232
Additional file 1. File format: PDF. Title: Supporting Information for “Evalua-	1233
tion Protocol, Checkpointing, and Calibration Shape Apparent Multi-Task Gains in	1234
ADMET Prediction”. Description: comparison with prior studies, complete endpoint-	1235
level results, calibration diagnostics, checkpoint-selection sensitivity, downsampling	1236
analyses, mechanistic controls, decision-level analyses, cross-architecture sensitivity,	1237
chemical-space diagnostics, and development history.	1238
	1239
	1240
	1241
	1242

References

- Jiang W, Zhang L, et al. MolP-PC: a multi-view fusion and multi-task learning framework for drug ADMET property prediction. Chinese Journal of Natural Medicines. 2025;[https://doi.org/10.1016/S1875-5364\(25\)60945-9](https://doi.org/10.1016/S1875-5364(25)60945-9).
- Shahid S, Maity D, Chakrabarty S. MTAN-ADMET: A Multi-Task Adaptive Neural Network for Efficient and Accurate Prediction of ADMET Properties. ChemRxiv. 2025;<https://doi.org/10.26434/chemrxiv-2025-zhrsk>.
- Fooladi M, et al. Evaluating Machine Learning Models for Molecular Property Prediction: Performance and Robustness on Out-of-Distribution Data. Journal of Chemical Information and Modeling. 2025;65(20):9871–9891. <https://doi.org/10.1021/acs.jcim.5c00475>.
- Guo J, et al. UMAP-based clustering split for rigorous evaluation of AI models for virtual screening on cancer cell lines. Journal of Cheminformatics. 2025;17:94. <https://doi.org/10.1186/s13321-025-01039-8>.
- Ahlmann-Eltze C, Huber W, Anders S. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. Nature Methods. 2025;22:1657–1661. <https://doi.org/10.1038/s41592-025-02772-6>.
- Wei Z, Wang Y, Gao Y, et al. Benchmarking algorithms for generalizable single-cell perturbation response prediction. Nature Methods. 2026;23:451–464. <https://doi.org/10.1038/s41592-025-02980-0>.
- Huang K, et al. Therapeutics Data Commons: Machine Learning Datasets and Tasks for Drug Discovery and Development. Nucleic Acids Research. 2021;49:D1153–D1160. <https://doi.org/10.1093/nar/gkaa977>.

Walter M, Borghardt JM, Humbeck L, Skalic M. Multi-Task ADME/PK Prediction at Industrial Scale: Leveraging Large and Diverse Experimental Datasets. Molecular Informatics. 2024;43(10):e202400079. https://doi.org/10.1002/minf.202400079 .	1289 1290 1291 1292 1293 1294
Kearnes S, et al. Modeling Industrial ADMET Data with Multitask Networks. arXiv preprint. 2016;1606.08793.	1295 1296 1297 1298
Zheng J, Guo C, Wang Z, Liu X. Scaffold splits hide structural-frontier failures in ADMET models. arXiv preprint. 2026;2607.10729.	1299 1300 1301 1302
Guo J, Ding S. Do Larger Models Really Win in Drug Discovery? arXiv preprint. 2026;2604.26498.	1303 1304 1305 1306
Wu S, Wang M, Yu L. Safe Multitask Molecular Graph Networks for Vapor Pressure and Odor Threshold Prediction. Chemometrics and Intelligent Laboratory Systems. 2026; https://doi.org/10.1016/j.chemolab.2026.104548 .	1307 1308 1309 1310 1311 1312
Mun V, Fazli S. CheMLT-F: multitask learning in biochemistry through transformer fusion. Journal of Cheminformatics. 2026; https://doi.org/10.1186/s13321-026-01199-1 .	1313 1314 1315 1316 1317 1318
RDKit contributors. RDKit: Open-source Cheminformatics Software. Open-source cheminformatics software. 2024; https://www.rdkit.org/ , version 2023.09.6, accessed 2026-08-03.	1319 1320 1321 1322 1323 1324
Bemis GW, Murcko MA. The properties of known drugs. 1. Molecular frameworks. Journal of Medicinal Chemistry. 1996;39(15):2887–2893. https://doi.org/10.1021/jm9602928 .	1325 1326 1327 1328 1329 1330 1331 1332 1333 1334

1335 Morgan HL. A Generation of Structural Fingerprints for Automated Classification and
1336
1337 Search of Chemical Structures. Journal of Chemical Documentation. 1965;5(2):107–
1338 113. <https://doi.org/10.1021/c160017a018>.
1339
1340
1341 Guo C, Pleiss G, Sun Y, Weinberger KQ. On Calibration of Modern Neural Networks.
1342 Proceedings of the 34th International Conference on Machine Learning (ICML).
1343 2017;p. 1321–1330.
1344
1345
1346
1347
1348
1349
1350
1351
1352
1353
1354
1355
1356
1357
1358
1359
1360
1361
1362
1363
1364
1365
1366
1367
1368
1369
1370
1371
1372
1373
1374
1375
1376
1377
1378
1379
1380